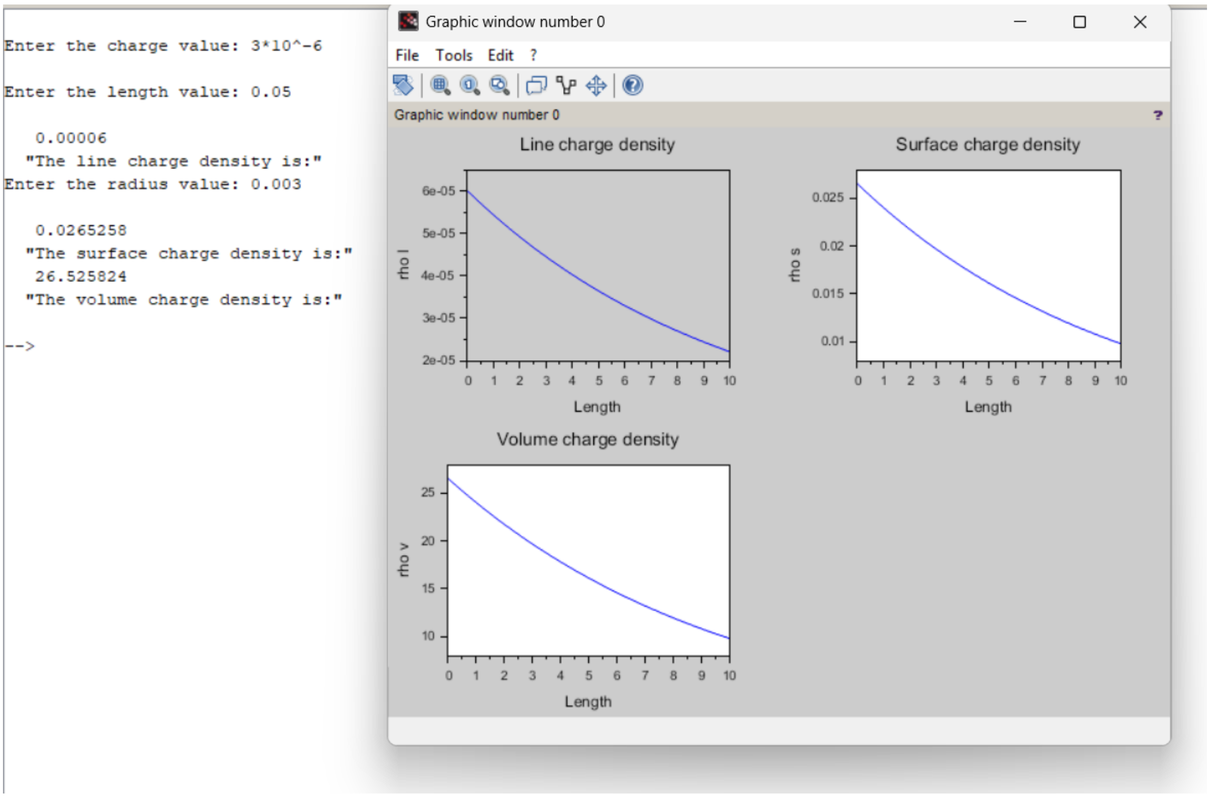
Exp. No: 4 Analysis of Continuous Charge Distribution in electric field. To find the (i) line charge density, (ii) Surface charge density and (iii) Volume charge density for any specific (a) line, (b) surface or (c) volume.

Program:

exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes

```
clc;
clear;
q = input("Enter the charge value: ");
l = input("Enter the length value: ");
// Line charge density
rl = q / l;
disp(rl, "The line charge density is:");
// Surface charge density
r = input("Enter the radius value: ");
rs = q / (4 * %pi * r^2);
disp(rs, "The surface charge density is:");
// Volume charge density
rv = q / ((4/3) * %pi * r^3);
disp(rv, "The volume charge density is:");
// Plotting
x = linspace(0, 10, 50);
y = rl * exp(-0.1 * x);
z = rs * exp(-0.1 * x);
v = rv * exp(-0.1 * x);
figure();
subplot(2,2,1);
plot(x,y);
title("Line charge density");
xlabel("Length");
ylabel("rho l");
subplot(2,2,2);
plot(x,z);
title("Surface charge density");
xlabel("Length");
ylabel("rho s");
subplot(2,2,3);
plot(x,v);
title("Volume charge density");
xlabel("Length");
ylabel("rho v");
```

Output;


Enter the charge value: 3*10^-6
Enter the length value: 0.05
0.00006
"The line charge density is:"
Enter the radius value: 0.003
0.0265258
"The surface charge density is:"
26.525824
"The volume charge density is:"
-->
Graphic window number 0
File Tools Edit ?
Graphic window number 0
Line charge density
rho l
Length
Surface charge density
rho s
Length
Volume charge density
rho v
Length

Calculation:

EXP :- 4

Enter the charge value (q) :- 3×10^{-6}

Enter the Length value (L) :- 5 cm = 0.05 m

Enter the radius value (R) = 3 mm = 0.003 m

A) given,

• Line charge density

$q = 3\times10^{-6}$

$L = 0.05$

$R = 0.03$

$$\boxed{\rho_L = \frac{q}{L}} = \frac{3\times10^{-6}}{0.05} = 6.0\times10^{-5}\ C/m$$

$\therefore 6.0\times10^{-5} \Rightarrow 0.00006$ //

• Surface charge density

O/P

$$\boxed{\rho_S = \frac{q}{4\pi R^2}} = \frac{3\times10^{-6}}{4\pi(0.003)^2}$$

$\rho_S = 2.65\times10^{-2}\ C/m^2$

$\therefore 2.65\times10^{-2} \Rightarrow 0.0265393$ //

• Volume charge density.

$$\boxed{\rho_V = \frac{q}{\frac{4}{3}\pi R^2}} = \frac{3\times10^{-6}}{\frac{4}{3}\pi\ (0.003)^2}$$

$\rho_V = 0.0266\ C/m^2$

$\therefore 0.266\ C/m^2 \Rightarrow 26.605793$ mm //

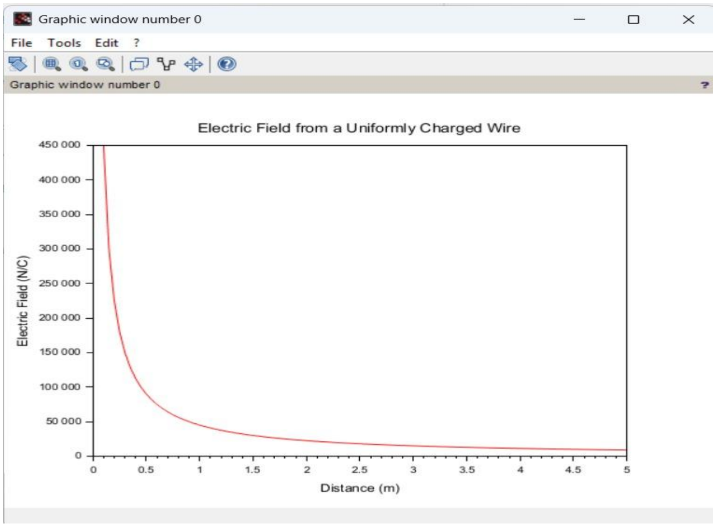
Case 1:

A long wire of length L=10m carries a uniform charge of Q=50μC. The goal is to calculate the line charge density λ and determine how the electric field behaves as a function of distance from the wire.

Code:

```
// Parameters
clc;
L = 10; // Length of the wire (m)
Q = 50e-6; // Total charge on the wire (C)

// Calculate line charge density
lambda = Q / L; // Line charge density (C/m)
disp("Line charge density (lambda) = " + string(lambda) + " C/m");

// Distance range from the wire for electric field calculation
r = linspace(0.1, 5, 100); // Distance (m)

// Electric field calculation using Coulomb's Law for a line charge
ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
E = (ke * lambda) ./ r; // Electric field (N/C)

// Plot the electric field as a function of distance
plot(r, E, "r");
xlabel("Distance (m)");
ylabel("Electric Field (N/C)");
title("Electric Field from a Uniformly Charged Wire");
```

Output:



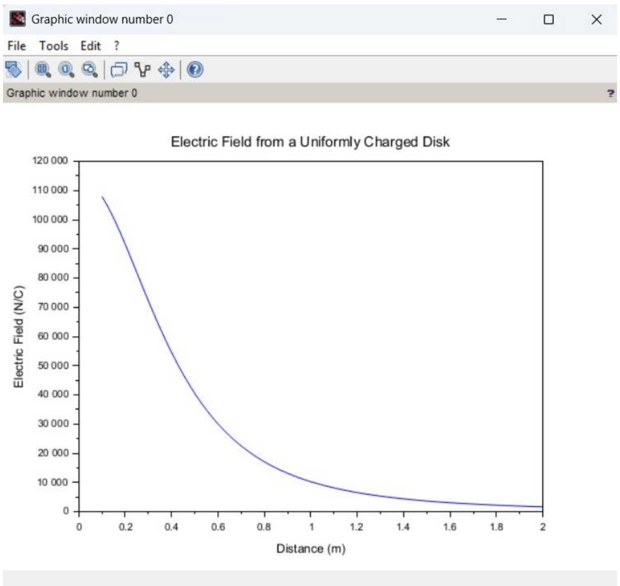
Case 2:

Given a uniformly charged disk of radius R=0.5m with total charge Q=10μC, calculate the surface charge density and plot the electric field along the axis of the disk. Explain the behavior of the electric field as a function of distance from the center of the disk.

Code:

```
// Parameters
clc;
R = 0.5; // Radius of the disk (m)
Q = 10e-6; // Total charge on the disk (C)

// Calculate surface charge density
A = %pi * R^2; // Area of the disk (m^2)
sigma = Q / A; // Surface charge density (C/m^2)
disp("Surface charge density (sigma) = " + string(sigma) + " C/m^2");

// Distance from the center of the disk to calculate electric field
z = linspace(0.1, 2, 100); // Distance (m)

// Electric field on the axis of the disk
ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
E_disk = (ke * sigma * R^2) ./ (2 * (z.^2 + R^2).^(3/2)); // Electric field (N/C)

// Plot the electric field as a function of distance
plot(z, E_disk, "b");
xlabel("Distance (m)");
ylabel("Electric Field (N/C)");
title("Electric Field from a Uniformly Charged Disk");
```

Output:



Case 3:

Given a uniformly charged sphere of radius R=2m with total charge Q=20μC, calculate the volume charge density and plot the electric field both inside and outside the sphere. Explain the difference in electric field behavior for points inside and outside the sphere.

Code:

```
// Parameters
clc;
R = 2; // Radius of the sphere (m)
Q = 20e-6; // Total charge (C)

// Calculate volume charge density
V = (4/3) * %pi * R^3; // Volume of the sphere (m^3)
rho = Q / V; // Volume charge density (C/m^3)
disp("Volume charge density (rho) = " + string(rho) + " C/m^3");

// Distance from the center of the sphere
r = linspace(0.1, 3, 100); // Distance (m)

// Electric field inside and outside the sphere (using Gauss's Law)
ke = 8.99e9; // Coulomb's constant (N*m^2/C^2)
E_sphere_inside = (ke * rho * r) / 3; // Electric field inside the sphere (N
E_sphere_outside = (ke * Q) ./ r.^2; // Electric field outside the sphere (N

// Plot the electric field inside and outside the sphere
plot(r, E_sphere_inside, "g", r, E_sphere_outside, "b");
xlabel("Distance (m)");
ylabel("Electric Field (N/C)");
title("Electric Field from a Uniformly Charged Sphere");
legend("Inside", "Outside");
```

Output: